Huawei Access Node IND-ACC-58 Optical Transport Failure SOP

Node: IND-ACC-58 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_4876

Description

The failure of the Optical Transport domain on the Huawei Access Node IND-ACC-58 at the Indiranagar Exchange site is primarily attributed to the degradation of SFP/XFP components, fiber cuts, or Rx/Tx power loss. This is due to the inherent limitations of the optical transmission technology employed by Huawei's Access Node equipment, which relies on the optical signal quality to ensure reliable data transmission. In the event of a fiber cut or SFP/XFP degradation, the optical signal quality is compromised, resulting in the failure of the Optical Transport domain. Furthermore, the Huawei Access Node equipment utilizes a complex interplay of protocols, including OTN (Optical Transport Network) and ODU (Optical Data Unit), to manage and transmit optical signals. Any issues with these protocols or the underlying hardware can lead to Optical Transport failures. Additionally, the physical environment of the Indiranagar Exchange site, including factors such as temperature, humidity, and vibration, can also contribute to Optical Transport failures.

Pre-requisite Checks

- Verify that the Indiranagar Exchange site is accessible and that all necessary equipment and personnel are on site.

- Check the status of the redundant links and ensure that they are functioning properly.

- Perform a visual inspection of the fiber optic cables and SFP/XFP components for any signs of damage or degradation.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-transport optical-connector , Expected Output: Optical Connector Information: Connector Type: SFP, Connector Name: SFP-1, Connector Status: Normal, Description: This command displays the optical connector information, including the connector type, name, and status. It helps to identify if the SFP/XFP component is functioning properly.

Command: display optical-transport fiber optic link, Expected Output: Fiber Optic Link Information: Link Name: FO-LINK-1, Link Status:
```

```
Normal, Link Length: 10km, Description: This command displays the
fiber optic link information, including the link name, status, and
length. It helps to identify if the fiber link is functioning
properly.

Command: display optical-transport optical-signal , Expected Output:
Optical Signal Information: Signal Type: OTU4, Signal Status: Normal,
Signal Quality: Good, Description: This command displays the optical
signal information, including the signal type, status, and quality. It
helps to identify if the optical signal is functioning properly.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP component, Command: replace optical-transport optical-connector SFP-1, Description: This command replaces the SFP/XFP component. Make sure to follow the proper procedures for replacing the component, including powering down the equipment and ensuring the new component is properly seated.

Step 2: Step: Tighten the fiber optic connector, Command: set optical-transport fiber optic link FO-LINK-1 connector-type tight, Description: This command tightens the fiber optic connector. Make sure to follow the proper procedures for tightening the connector, including ensuring it is properly seated and secured.

Step 3: Step: Adjust the optical signal power, Command: set optical-transport optical-signal OTU4 signal-power 0dBm, Description: This command adjusts the optical signal power. Make sure to follow the proper procedures for adjusting the signal power, including ensuring it is within the recommended range.

Post-Check Verification

```
Command: display optical-transport optical-connector , Expected
Output: Optical Connector Information: Connector Type: SFP, Connector
Name: SFP-1, Connector Status: Normal, Description: This command
verifies that the SFP/XFP component is functioning properly.

Command: display optical-transport fiber optic link, Expected Output:
Fiber Optic Link Information: Link Name: FO-LINK-1, Link Status:
Normal, Link Length: 10km, Description: This command verifies that the
fiber link is functioning properly.

Command: display optical-transport optical-signal , Expected Output:
Optical Signal Information: Signal Type: OTU4, Signal Status: Normal,
Signal Quality: Good, Description: This command verifies that the
optical signal is functioning properly.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP component, Command: undo replace optical-transport optical-connector SFP-1, Description: This command restores the original SFP/XFP component. Make sure to follow the proper procedures for restoring the component, including powering down the equipment and ensuring the original component is properly seated.

- Step: Loosen the fiber optic connector, Command: undo set optical-transport fiber optic link FO-LINK-1 connector-type tight, Description: This command loosens the fiber optic connector. Make sure to follow the proper procedures for loosening the connector, including ensuring it is properly seated and secured.

- Step: Reset the optical signal power, Command: undo set optical-transport optical-signal OTU4 signal-power 0dBm, Description: This command resets the optical signal power. Make sure to follow the proper procedures for resetting the signal power, including ensuring it is within the recommended range.